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Semester 2, 2026
Tutorial Week 3

COMP3670 / COMP6670: Introduction to Machine Learning

Question 1 **Which features are worth recording?**

A bank records, for every applicant in its population, two binary features and one binary outcome:

$$X_1 = \mathbb{1}\{\text{high income}\}, \qquad X_2 = \mathbb{1}\{\text{previous default}\}, \qquad Y = \mathbb{1}\{\text{the applicant defaults}\}.$$

The joint distribution. The vector (X_1, X_2, Y) consists of three binary random variables, so it can take $2^3 = 8$ different values. Its **joint distribution** is the function which assigns to each of those eight triples the proportion of the population having exactly those values:

$$p(x_1, x_2, y) \;=\; \mathbb{P}\big(X_1 = x_1,\; X_2 = x_2,\; Y = y\big).$$

For the population this bank serves the joint distribution is given by the table below.

x_1	x_2	y	$p(x_1, x_2, y)$
0	0	0	0.24
0	0	1	0.06
0	1	0	0.06
0	1	1	0.14
1	0	0	0.36
1	0	1	0.04
1	1	0	0.04
1	1	1	0.06
			1.00

Read the third row, for instance, as: 6% *of the applicants have a low income, have defaulted before, and do not default now.* Every applicant falls into exactly one of the eight rows, so the eight proportions are non-negative and add up to 1, as any distribution must. The joint distribution contains *everything* there is to know about this population: every probability you are asked for below is computed from it.

Marginal distributions. The **marginal** distribution of one variable (or of a group of variables) is its distribution on its own, ignoring the others. It is obtained by **summing the joint distribution over the variables we are ignoring**. For instance,

$$\mathbb{P}(Y = y) \;=\; \sum_{x_1 \in \{0,1\}} \; \sum_{x_2 \in \{0,1\}} p(x_1, x_2, y), \qquad \mathbb{P}(X_1 = x_1) \;=\; \sum_{x_2 \in \{0,1\}} \; \sum_{y \in \{0,1\}} p(x_1, x_2, y).$$

The reason is just that of adding disjoint groups: the applicants who default are exactly those in the rows with $y = 1$, so their total proportion is the sum of those rows, whatever the features

happen to be. Conditional probabilities are then obtained, as in the lecture, by **dividing the joint by the marginal**.

Throughout, use the **binary loss** $\ell((\mathbf{x}, y), h) = \mathbb{1}\{h(\mathbf{x}) \neq y\}$, so that the out-of-sample error of a rule h is $E_{\text{out}}(h) = \mathbb{P}(h(\mathbf{X}) \neq Y)$.

Independence: We say that two discrete random variables X and Y are *independent* if

$$\mathbb{P}(Y = y|X = x) = \mathbb{P}(Y = y), \text{ or, equivalently, } \mathbb{P}(Y = y, X = x) = \mathbb{P}(Y = y)\mathbb{P}(X = x)$$

for all possible values y and x that Y and X can take, respectively. Being independent means that, knowing the value of X (conditioned on X), does not change the probability distribution of Y.
When things (i.e., events) are independent, the probability of they occurring together (i.e., joint probability), is just the product of the probabilities of they occurring by themselves (i.e., marginal probability).

(a) Compute the marginal distribution of Y, the marginal distribution of X_1 and the marginal distribution of X_2, by summing the joint as above. Compute also the joint distribution of the features alone, $\mathbb{P}(X_1 = x_1, X_2 = x_2)$, for the four combinations of (x_1, x_2).

(b) If the bank does not record features of applicants at all, it must predict the same credit outcome for everyone: either default (1) or repay (0). That is, it must use a constant rule $h(x_1, x_2)$ that assigns the same value to every (x_1, x_2). What is the best constant rule, and what is its out-of-sample error? Call this the **baseline error**.

(c) Compute $\mathbb{P}(Y = 1 \mid X_1 = 0)$ and $\mathbb{P}(Y = 1 \mid X_1 = 1)$. Are X_1 and Y independent?

(d) Suppose the bank records **both** features. Compute

$$\eta(x_1, x_2) = \mathbb{P}(Y = 1 \mid X_1 = x_1, X_2 = x_2)$$

for all four combinations of (x_1, x_2), write down the target function $f^\star$, and compute $E_{\text{out}}(f^\star)$.

(e) **(Optional)** Suppose the bank records **only** X_1. Compute the target function based on X_1 alone and its out-of-sample error.

(f) **(Optional)** Suppose instead the bank records **only** X_2. Compute the target function based on X_2 alone and its out-of-sample error.

(g) **(Optional)** Compare (b), (d), (e) and (f). Two things should look surprising; explain both:

(i) recording X_2 alone is as good as recording both features;

(ii) X_1 carries genuine information about Y, by (c), and yet recording X_1 alone is worth *nothing* at all.

Solution.
Everything below is obtained from the eight numbers of the joint distribution, by summing (for marginals) and dividing (for conditionals).

(a) **Marginal of Y**: add the rows with $y = 1$, and the rows with $y = 0$:

$$\mathbb{P}(Y = 1) = 0.06 + 0.14 + 0.04 + 0.06 = 0.30,$$
$$\mathbb{P}(Y = 0) = 0.24 + 0.06 + 0.36 + 0.04 = 0.70.$$

They add to 1, as a distribution must.

Marginal of X_1: add the four rows with $x_1 = 0$, and the four with $x_1 = 1$:

$$\mathbb{P}(X_1 = 0) = 0.24{+}0.06{+}0.06{+}0.14 = 0.50, \qquad \mathbb{P}(X_1 = 1) = 0.36{+}0.04{+}0.04{+}0.06 = 0.50.$$

Marginal of X_2: add the four rows with $x_2 = 0$, and the four with $x_2 = 1$:

$$\mathbb{P}(X_2 = 0) = 0.24{+}0.06{+}0.36{+}0.04 = 0.70, \qquad \mathbb{P}(X_2 = 1) = 0.06{+}0.14{+}0.04{+}0.06 = 0.30.$$

Joint of the features alone: here we sum over y only, two rows at a time, for fixed values of X_1, X_2:

$$\mathbb{P}(X_1 = 0, X_2 = 0) = 0.24 + 0.06 = 0.30, \quad \mathbb{P}(X_1 = 0, X_2 = 1) = 0.06 + 0.14 = 0.20,$$
$$\mathbb{P}(X_1 = 1, X_2 = 0) = 0.36 + 0.04 = 0.40, \quad \mathbb{P}(X_1 = 1, X_2 = 1) = 0.04 + 0.06 = 0.10.$$

These four also add to 1: they are a distribution over the four kinds of applicant.

(b) With no features the only rules available are $h \equiv 0$ and $h \equiv 1$, with errors $\mathbb{P}(Y = 1) = 0.30$ and $\mathbb{P}(Y = 0) = 0.70$ respectively. The best constant rule is $h \equiv 0$ ("predict that nobody defaults"), with

$$\textbf{baseline error} \;=\; 0.30.$$

(c) Divide the joint by the marginal. The applicants with $X_1 = 0$ and $Y = 1$ are the rows $(0, 0, 1)$ and $(0, 1, 1)$, so

$$\mathbb{P}(Y = 1 \mid X_1 = 0) = \frac{0.06 + 0.14}{0.50} = \frac{0.20}{0.50} = 0.4,$$
$$\mathbb{P}(Y = 1 \mid X_1 = 1) = \frac{0.04 + 0.06}{0.50} = \frac{0.10}{0.50} = 0.2.$$

Neither equals the marginal $\mathbb{P}(Y = 1) = 0.30$, so X_1 and Y are **not independent**: knowing X_1 does change the distribution of Y. High-income applicants default half as often as low-income ones.

(d) Divide each $y = 1$ row by the corresponding feature probability from (a):

$$\eta(0,0) = \tfrac{0.06}{0.30} = 0.2, \quad \eta(0,1) = \tfrac{0.14}{0.20} = 0.7, \quad \eta(1,0) = \tfrac{0.04}{0.40} = 0.1, \quad \eta(1,1) = \tfrac{0.06}{0.10} = 0.6.$$

The target function predicts the more probable label, $f^\star = \mathbb{1}\{\eta > 1/2\}$:

$$f^\star(0,0) = 0, \quad f^\star(0,1) = 1, \quad f^\star(1,0) = 0, \quad f^\star(1,1) = 1.$$

The error left at each combination is the probability of the label given up, $\min\{\eta, 1 - \eta\}$, namely 0.2, 0.3, 0.1, 0.4. Averaging these over the features, weighted by $\mathbb{P}(X_1 = x_1, X_2 = x_2)$ from (a),

$$E_{\text{out}}(f^\star) = 0.30(0.2) + 0.20(0.3) + 0.40(0.1) + 0.10(0.4) = 0.06 + 0.06 + 0.04 + 0.04 = \mathbf{0.20}.$$

Equivalently: $f^\star$ misclassifies exactly the applicants on the minority side of each combination, a proportion $0.06 + 0.06 + 0.04 + 0.04 = 0.20$ of the population.

(e) From (c), $\eta_1(0) = 0.4$ and $\eta_1(1) = 0.2$. *Both are below* $1/2$, so the target function based on X_1 alone is

$$f_1^\star(0) = 0, \qquad f_1^\star(1) = 0,$$

i.e. the constant rule $h \equiv 0$. Its error is

$$E_{\text{out}}(f_1^\star) = 0.50(0.4) + 0.50(0.2) = 0.20 + 0.10 = \mathbf{0.30},$$

exactly the baseline of (b).

(f) Now condition on X_2, using $\mathbb{P}(X_2 = 0) = 0.70$ and $\mathbb{P}(X_2 = 1) = 0.30$ from (a):

$$\eta_2(0) = \frac{0.06 + 0.04}{0.70} = \frac{0.10}{0.70} = \frac{1}{7}, \qquad \eta_2(1) = \frac{0.14 + 0.06}{0.30} = \frac{0.20}{0.30} = \frac{2}{3}.$$

Hence $f_2^\star(0) = 0$ and $f_2^\star(1) = 1$, with error

$$E_{\text{out}}(f_2^\star) = 0.70 \cdot \tfrac{1}{7} + 0.30 \cdot \tfrac{1}{3} = 0.10 + 0.10 = \mathbf{0.20}.$$

(g) Collecting the four numbers:

features recorded	best achievable error
none (baseline error)	0.30
X_1 only	0.30
X_2 only	0.20
both X_1, X_2	0.20

(i) Look at $f^\star$ in (d): it outputs 0 whenever $X_2 = 0$ (both combinations) and 1 whenever $X_2 = 1$ (both combinations). **The optimal rule never actually consults X_1.** Knowing X_1 does change η (compare $\eta(0,1) = 0.7$ with $\eta(1,1) = 0.6$) but it never moves η across the threshold $1/2$, so it never changes the *decision*. Since the rule of (d) and the rule of (f) are the same function, they must have the same error. Recording X_1 buys the bank nothing once X_2 is on the form.

(ii) By (c), X_1 genuinely carries information: η_1 moves from 0.4 to 0.2 as X_1 goes from 0 to 1. But under the binary loss all that matters is which side of $1/2$ we land on, and 0.4 and 0.2 are on the *same* side. So the best rule that uses X_1 alone is forced to give the same answer for both groups, and collapses to the constant baseline.

The moral. "X is informative about Y" and "X is useful for predicting Y" are different statements. Under the binary loss a feature pays off only when it moves η *across* $1/2$ for some applicants. A feature can be strongly associated with the outcome and still be worthless on its own, and, as in (i), a feature can be informative and still be redundant next to a better one.

Question 2 The smart algorithm and the crazy algorithm

What the bank can actually compute. Every number in Question 1 came out of the *population* joint distribution: the eight proportions were simply handed to us. A real bank does not have them. All it ever sees is a **sample** $\mathcal{D}_N$ of N applicants drawn from the population, and from a sample it can:

compute $E_{\text{in}}(h)$ for any rule h it writes down, by simply counting its mistakes on those N applicants.

But it can **never** compute $E_{\text{out}}(h)$, or the target function $f^\star$, or its out-of-sample error, exactly: those are properties of the population distribution, and the sample is not the population. From here on, the bank must choose a hypothesis using only what a sample shows it.

Suppose moreover that the bank records **no features at all**, so that the only rules available are the two constants

$$\mathcal{H} = \{h_0, h_1\}, \qquad h_0 \equiv 0 \ \ (\text{“nobody defaults”}), \qquad h_1 \equiv 1 \ \ (\text{“everybody defaults”}).$$

It must choose between h_0 and h_1 using only a sample of $N = 5$ applicants drawn independently from the population. Since no features are recorded, all the sample contains is the outcomes, so here we write $\mathcal{D}_N = \{Y_1, \ldots, Y_N\}$. We compare two learning algorithms:

S (smart) returns the hypothesis with the *smaller* in-sample error on $\mathcal{D}_N$;

C (crazy) returns the other one: the hypothesis with *greater* in-sample error on $\mathcal{D}_N$.

Let K denote the number of applicants in the sample who default.

Counting the possible samples. Write down what happens to each of the five applicants in order, using D if the applicant defaults and N if not. A possible sample is then a sequence of five letters, such as NDNND. Suppose each applicant defaults with probability π, independently of the others.

Since the five applicants are independent, recall from the lecture that the probability of *one particular* sequence is the product of five probabilities, one per applicant: π for each D and $1-\pi$ for each N. For example

$$\mathbb{P}(\mathsf{NDNND}) = (1-\pi)\,\pi\,(1-\pi)\,(1-\pi)\,\pi = \pi^2(1-\pi)^3.$$

So *every* sequence with exactly k letters D has the same probability $\pi^k(1-\pi)^{5-k}$, no matter where the D's sit. To find $\mathbb{P}(K = k)$, the probability of k defaults in a sample of size 5, we therefore only have to count how many sequences have exactly k letters D, and multiply:

$k = 0$: only NNNNN. That is 1 sequence.

$k = 1$: the single D can sit in position 1, 2, 3, 4 or 5. That is 5 sequences.

$k = 2$: list the pairs of positions holding the two D's: $12, 13, 14, 15$, then $23, 24, 25$, then $34, 35$, then 45. That is $4 + 3 + 2 + 1 = 10$ sequences.

$k = 3$: choosing which three applicants default is the same as choosing which two do *not*, so there are again 10 sequences.

$k = 4$: choosing which four default is the same as choosing which one does not, so 5.

$k = 5$: only DDDDD. That is 1 sequence.

As a check, the six counts add to $1 + 5 + 10 + 10 + 5 + 1 = 32$, which is the total number of five-letter sequences: two choices for each of the five applicants, $2 \times 2 \times 2 \times 2 \times 2 = 32$. Multiplying each count by the probability of one such sequence gives the distribution of K, called the **binomial distribution** with parameters 5 and π:

k	how many sequences	probability of each	$\mathbb{P}(K=k)$
0	1	$(1-\pi)^5$	$(1-\pi)^5$
1	5	$\pi(1-\pi)^4$	$5\,\pi(1-\pi)^4$
2	10	$\pi^2(1-\pi)^3$	$10\,\pi^2(1-\pi)^3$
3	10	$\pi^3(1-\pi)^2$	$10\,\pi^3(1-\pi)^2$
4	5	$\pi^4(1-\pi)$	$5\,\pi^4(1-\pi)$
5	1	π^5	π^5

(a) Using your answers to Question 1, write down $E_{\text{out}}(h_0)$ and $E_{\text{out}}(h_1)$. Which of the two is the better hypothesis? Could the bank determine this from its sample alone?

(b) Write $E_{\text{in}}(h_0)$ and $E_{\text{in}}(h_1)$ in terms of K. Which hypothesis does S return, as a function of K? Why can there never be a tie?

(c) What is the value of π for this bank? Using the table, write down $\mathbb{P}(K=0)$, $\mathbb{P}(K=1)$ and $\mathbb{P}(K=2)$ as numbers. [**Note**: you may use the calculator on your phone.]

(d) Compute the probability that S returns the better hypothesis, and hence the probability that the crazy algorithm beats the smart one.

(e) **(Optional)** Is S *guaranteed* to return the better hypothesis? Is there any sample size N for which it would be?

Solution.

(a) A constant rule is wrong exactly on the applicants of the other class, so by the marginal of Y computed in Question 1(a),

$$E_{\text{out}}(h_0) = \mathbb{P}(Y=1) = 0.30, \qquad E_{\text{out}}(h_1) = \mathbb{P}(Y=0) = 0.70.$$

So h_0 is the **better** hypothesis, and it is the baseline of Question 1(b). But this comparison uses the population distribution, which the bank does not have: from the sample it can only compute E_{in}, never E_{out}. It cannot *know* that h_0 is better: it can only guess, and that is exactly what S does.

(b) h_0 predicts 0 for everybody, so it is wrong precisely on the K applicants in the sample who default; h_1 is wrong on the other $5-K$. Hence

$$E_{\text{in}}(h_0) = \frac{K}{5}, \qquad E_{\text{in}}(h_1) = \frac{5-K}{5}.$$

Therefore

$$\text{S returns } h_0 \iff K < 5-K \iff K < \tfrac{5}{2} \iff K \le 2,$$

and S returns h_1 when $K \ge 3$. A tie would require $K = 5-K$, that is $K = 2.5$, which is impossible for a whole number: with $N=5$ **odd** the two hypotheses can never have the same in-sample error.

(c) An applicant drawn at random defaults with probability $\mathbb{P}(Y = 1) = 0.3$, so $\pi = 0.3$ and $1 - \pi = 0.7$. Reading the first three rows of the table:

$$\begin{aligned}
\mathbb{P}(K = 0) &= (0.7)^5 = 0.16807, \\
\mathbb{P}(K = 1) &= 5\,(0.3)(0.7)^4 = 1.5 \times 0.2401 = 0.36015, \\
\mathbb{P}(K = 2) &= 10\,(0.3)^2(0.7)^3 = 0.9 \times 0.343 = 0.30870.
\end{aligned}$$

(d) By (a) the better hypothesis is h_0, and by (b) S returns it exactly when $K \leq 2$. Adding the three probabilities of (c),

$$\mathbb{P}(\text{S beats C}) = \mathbb{P}(K \leq 2) = 0.16807 + 0.36015 + 0.30870 = \mathbf{0.83692}.$$

The crazy algorithm wins with probability $1 - 0.83692 = \mathbf{0.16308}$, roughly one time in six. It wins whenever the five applicants drawn happen to include three or more defaulters, which makes the sample look as though defaulting were the majority outcome.

(e) **No, and no.** For *every* sample size the probability that more than half of the sample defaults is strictly positive; for instance, all N applicants drawn could be defaulters, a single sequence of probability $(0.3)^N > 0$. Whenever that happens S returns h_1, the worse hypothesis, and C wins. So no sample size makes S *certain* to be right; increasing N only drives that probability towards 0, never to 0.

The moral. Nothing here is guaranteed: a sample can always mislead. What the probabilistic view buys us is that the chance of being misled is something we may be able to *control*, and one we can shrink by collecting more data.